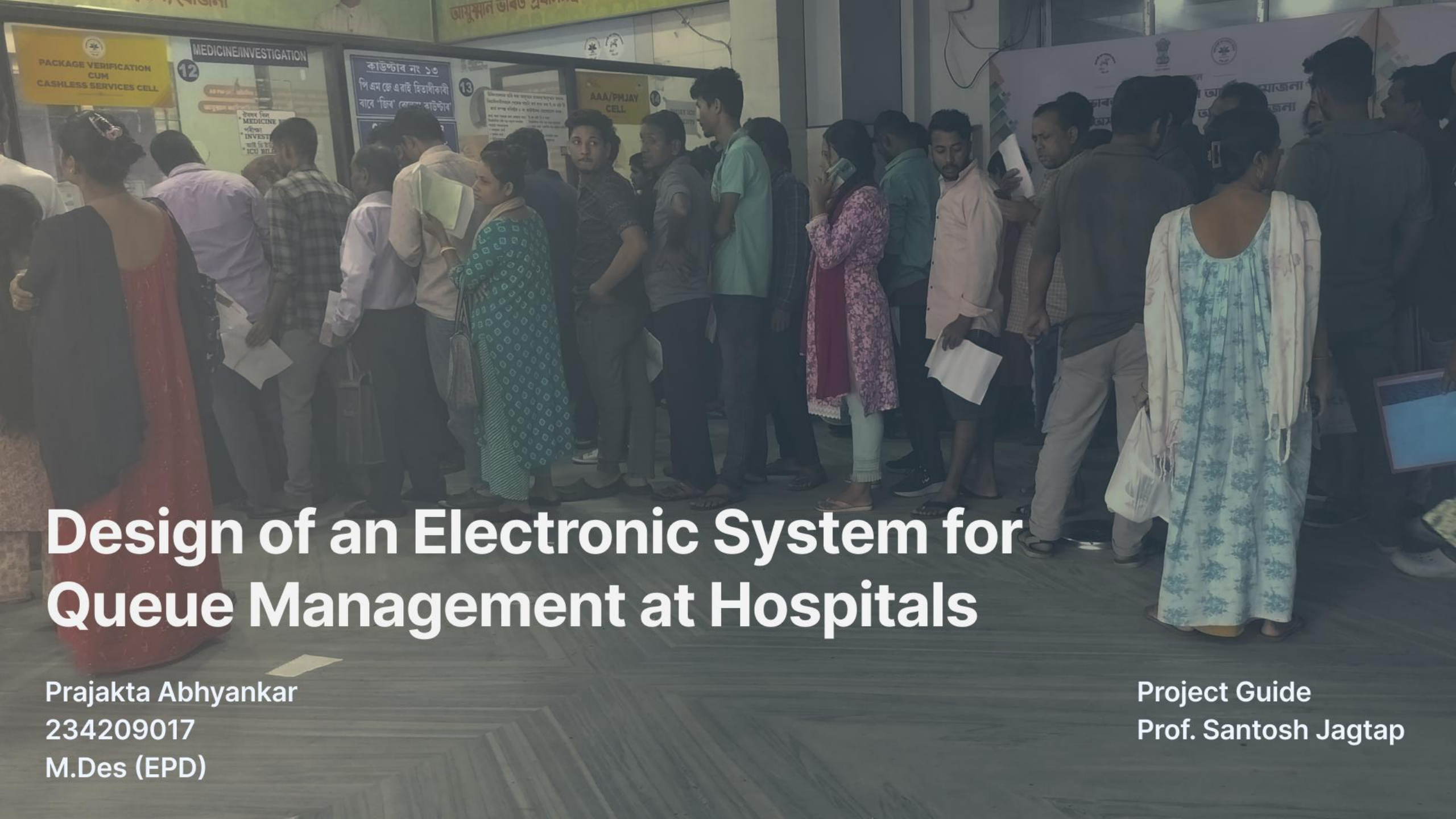


A photograph showing a long queue of people, mostly men, waiting at a hospital counter. The queue is formed in front of a counter with several service windows. Above the windows, there are signs in English and Bengali. The English signs include 'PACKAGE VERIFICATION CUM CASHLESS SERVICES CELL', 'MEDICINE/INVESTIGATION', and 'AAA/PMJAY CELL'. The Bengali signs include 'কাউন্টার নং ১৩' (Counter No. 13), 'পিএম জে এরই দিয়ারীকরী বারে 'জিবি' লেভেল কাউন্টার' (At the PM-Jay level, the counter for 'Jibi' level), and 'কাউন্টার নং ১৩' (Counter No. 13). The people in the queue are dressed in casual clothing, and some are holding papers. The background shows a building with a sign that says 'আমুসান ভারত প্রকল্প' (Amusana Bharat Project).

Design of an Electronic System for Queue Management at Hospitals

Prajakta Abhyankar
234209017
M.Des (EPD)

Project Guide
Prof. Santosh Jagtap

Overview

This project focuses on improving the queue management experience in hospitals by designing an electronic token-based system tailored for both government and private healthcare facilities. It aims to streamline patient registration, reduce waiting time confusion, and improve communication between patients, attendants, and hospital staff.

The system currently uses an RFID-based identification, coin motor haptics, mini thermal printing, and an LCD display, and simulates a real-time interaction flow for patients. Future upgrades include biometric access, mobile integration, cloud dashboards, and AI-powered queue prediction.

By conducting field research in both government and private hospitals, this project also identifies and addresses gaps in patient experience, attendant roles, and technology adoption, proposing a unified solution that bridges the gap between the two sectors.

Secondary Research

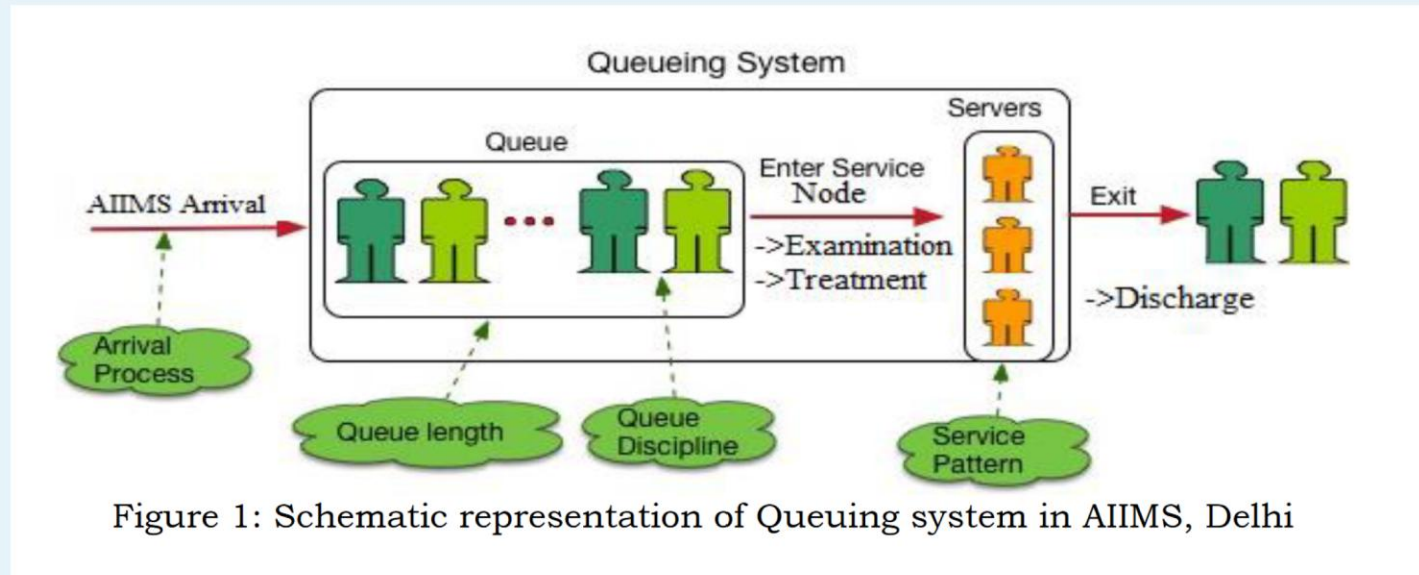
Research on AIIMS Delhi for their Queue Management System.

Key Insights

Waiting time: *The difference between when a patient arrived and when service started.*

Service Time: *Difference between the time a patient entered until when he/she exited the system/mean time in the system*

Length of Queue: *The length of the queue at a particular point in time (*



Statistics retrieved from the Paper

Table 2: Mean weekly data of AIIMS general OPD at each server

Days	No. of patients	Average waiting time in queue (mins)	Average service time (min)	Total time in system (min)
Monday	45	162	23	185
Tuesday	41	155	18	173
Wednesday	39	160	16	176
Thursday	41	158	20	178
Friday	35	145	22	167
Average	40	156	20	176

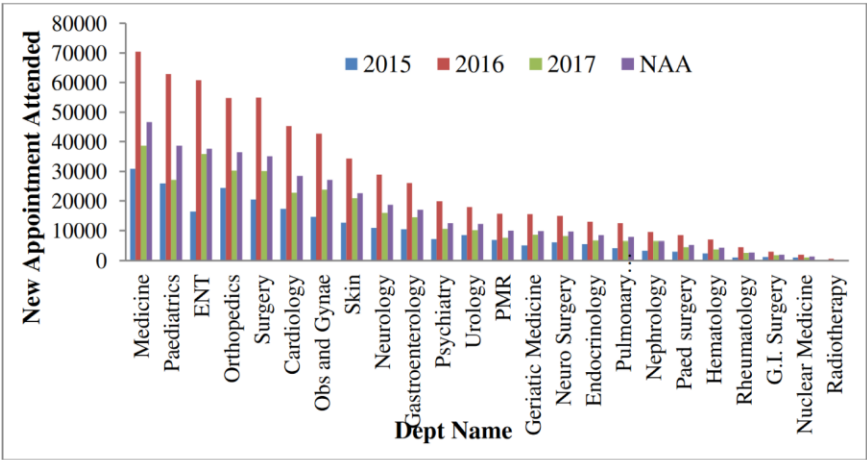


Figure 5: AIIMS Department wise visited/NAA patients (2015-2017)

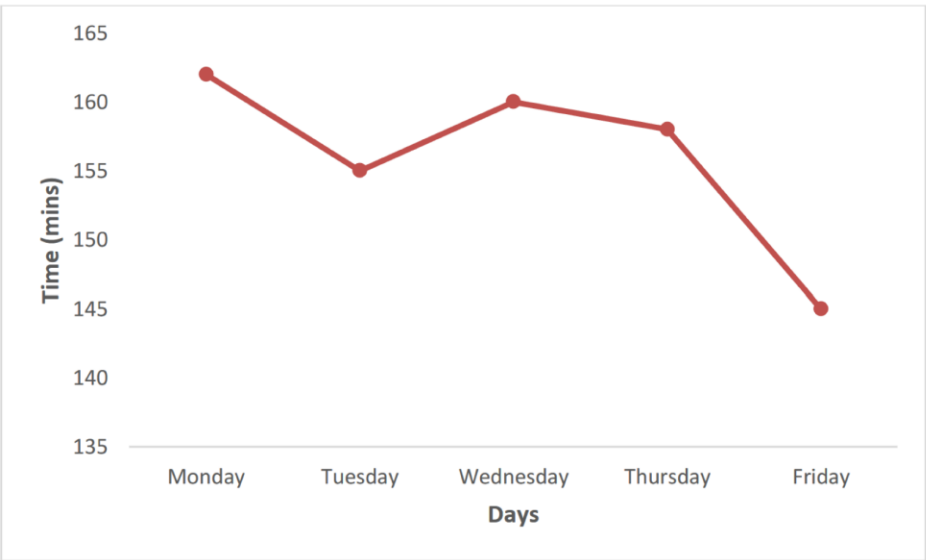


Figure 6: Average waiting time in queue of patients (in mins)

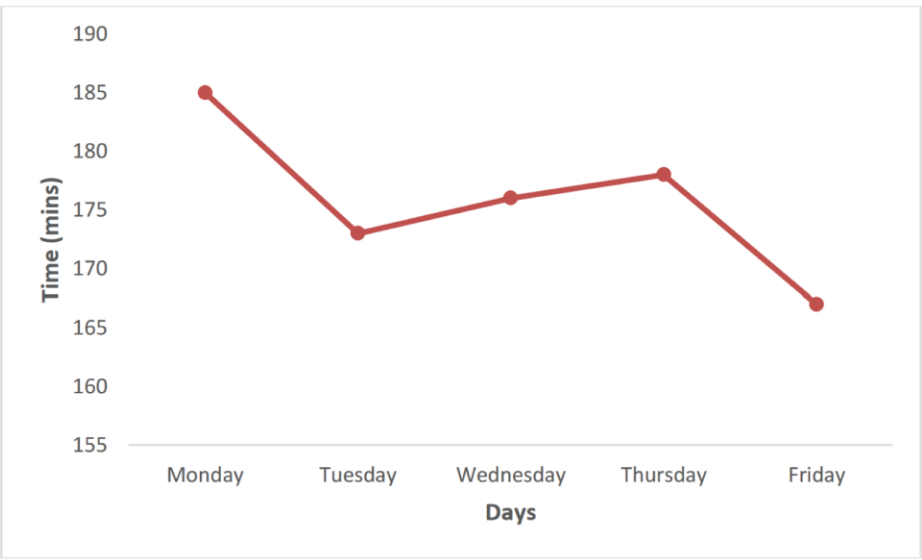
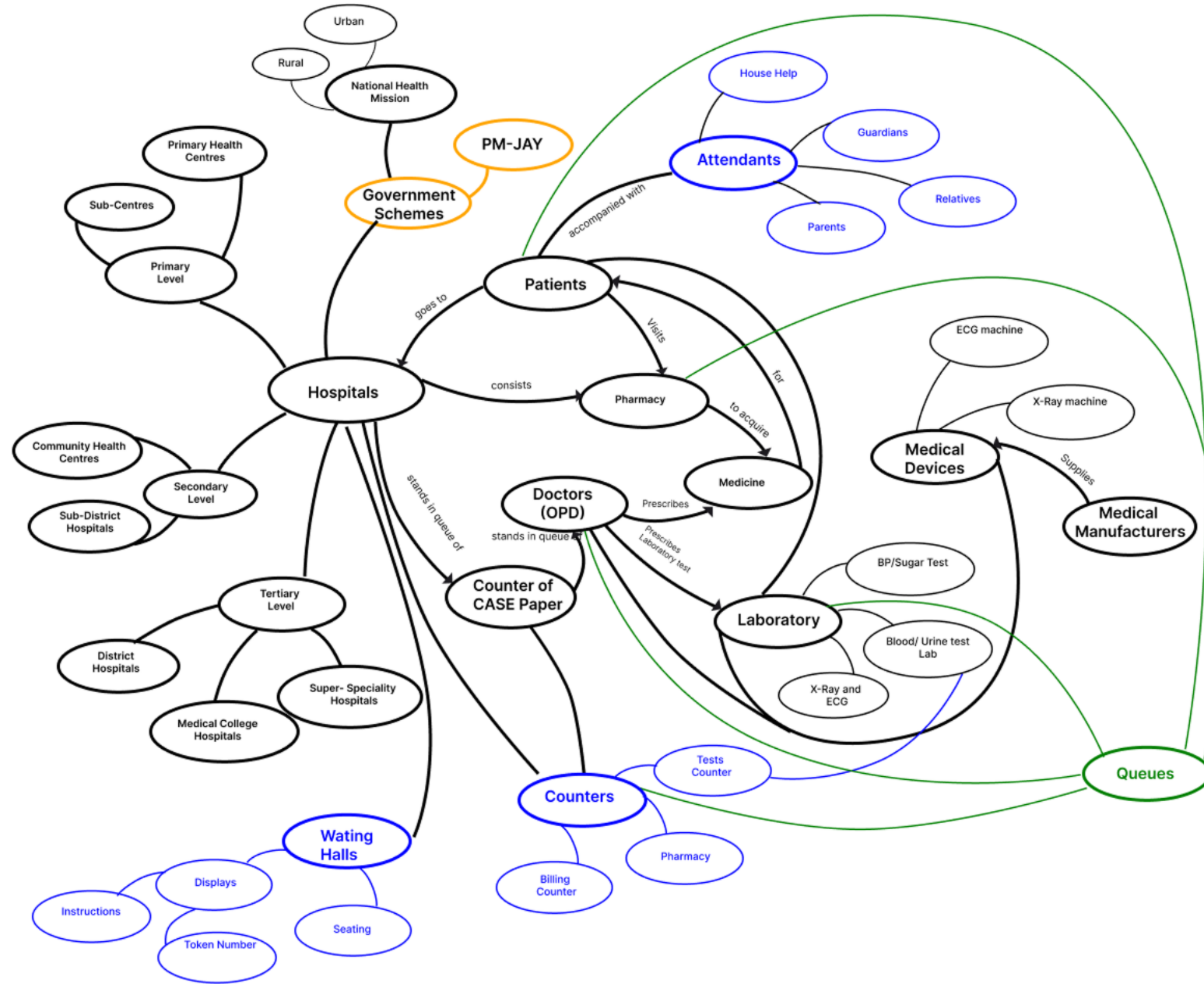


Figure 7: Total time spent by patient in hospital (in mins)

Understanding the Hospital as a System



Thematic Analysis

Blood Donation Procedure

Prajakta A

Blood donation counter

Prajakta A

Pink color receipts/slips for blood donation.

Prajakta A

Patients tend to submit the blood in the wrong counter as well.

Prajakta A

To donate blood , certain medical test are performed on the donar

Prajakta A

Admission of patient

Prajakta A

Blood donation process involves

Prajakta A

5-6 min of queue time

Prajakta A

Form filling 1-5 min

Prajakta A

Wait time approx 30 min

Prajakta A

5-10 min tests are done on the donar

Prajakta A

Have you eaten enough
slept well enough

Prajakta A

Transgender need to take HIV test as well

Prajakta A

Admitted on 16th nov, 10 pm

Prajakta A

They stayed here over night

Prajakta A

Patient and Attendant interviews

User Interview -3
17 th nov
Attendant - Lady with 2 kids
Patient- Kid's father aunt

User Interview -1
17 th nov
Jenne
Transgender

User Interview -5
21st Nov
Attendant- standing in the
Ayushman queue

User Interview -4
21st Nov
Patient-Ajay Das
25 yrs old

User Interview -2
17 th nov
Attendant - Daughter
Patient- Mother

User Interview -6
21st Nov
Patient- Hari Das
Debananth

User Interview -4
17 th nov
Attendant - Sister
Patient-brother

User Interview -7
21st Nov
Patient-

Govt Scheme

Holding the
ayushman
application form

Information of
patients
details : name,
age, address, etc

Disease Name
Disease ID code
number

The intestinal
operation got
done by
Ayushman
scheme.

Ayushman ID:
Disease code
Per day expense
limit e.g 2000rs

The medical bills
of accident is
done by
ayushman
scheme.

For one year,
One lakh rupees
of medical bills
are free.

PM-JAY
Ayushman
scheme

With ayushman
test cost is for
free

I was standing in ayushman
queue till 6pm now another
member is standing.

The ayushman
seal needs to be
there on the
receipts (Hypo)

There's a list of
diseases which
comes under the
scheme.

Ayushman
scheme is not for
normal fever and
cold

Ayushman card
is active and
available

An application
form needs to be
submitted on the
counter.

These diseases
have their own ID
number.

Ayushman card
is not active

There are
different counter
for opening and
closing

To activate the
card . They have
to stand in the
queue

There should be a
ayushman seal on
the receipt of
blood test.

After taking the
seal by standing
in the ayushman
queue.

For making blood
test free of cost
Ayushman seal
is required on the
receipt.

Observations

They are stressed and in this
chaos it is difficult for them to
navigate and process things
around in the hospital.

Registration counters are filled
with paper work

Few kids are
wearing masks

The patients and attendants gets
hungry by spending long hours in
the hospital.

One vendor outside the gate is
selling file folders and masks.

To protect themselves from
getting any infection

They stay in the hospital
premises overnight.

Vendors sell snacks to the
patients in the waiting hall

They roam around in circles
inside the hospitals navigating
where to go and what to do!

They are not clear
on where to go
exactly and what
to do!

Patient

Project A

Patient is facing liver problem caused by excessive drinking

Project A

Patient constantly keeps an eye on the attendant to see where he is standing in the queue

Project A

The patient rest in the hospital's corridor

Project A

Patient is suffering from stones in stomach

Project A

Patients and attendants are clueless

Project A

The patient has gone through an accident

Project A

Patients and attendants trust the Doctors to navigate them or tell them what to do next

Project A

patient facing gall bladder stone issue

Project A

OPD patient

Project A

Patient had an intestinal major operation

Project A

THIS patient didn't have an attendant

Project A

The patient is going to spent the night here in hospital for early morning next days appointment

Project A

The bone broken of one person.

Project A

The other person is also admitted

Project A

Aunt is facing Liver tumour problem got admitted on 13th nov

Project A

He was kept on hold for 7 days.

Project A

Worst case scenarios where the 40yrs old with mental health issue

Project A

Doctors

Project A

Doctors want the reports of the patient immediately for further diagnosis.

Project A

Only on thursday Gastrology Dr. is available

Project A

Doctors have specific weekly based duties

Project A

Their days of duty is fixed on that day of the week ,

Project A

He is accompanied by one more person

Project A

Today is Thursday and appointment with Dr is on next day i.e friday

Project A

After 10-15 days , the patient is called for a checkup

Project A

This might cause delay in diagnosis that's why all this precautionary measures.

Project A

They will have to stand again queue early in morning for appointment

Project A

This is bcz of doctor's availability on that day of the week else thwy have have to come nxt week

Project A

Attendant

Project A

Lady attendant Talking to me and meanwhile handling the kids to wear change of clothes brought in the bag.

Project A

The father is standing in the queue.

Project A

Daily commuting to the hospital on a regular basis

Project A

Lady stood in the queue of blood collection counter

Project A

Lady is also 6 month pregnant

Project A

Rest all the queues father stood in line

Project A

Having a notion of nothing will change

Project A

Attendant sits on the floor while standing in queue for long hours

Project A

one more attendant accompanying the patient

Project A

They recognize by looking at the apron or doctor's coat

Project A

They came to donate blood

Project A



Two girls both are trans

Project A

total Attendants -3

Project A

Standing alone in the queue exhausted

Project A

Patient's mother admitted the patient

Project A

Stood in queue for long hours

Project A

TRAVEL of Patients

Project A

Travelled from Tamlpur which takes 2 hrs to reach hospital.

Project A

They came here from Narangi, which takes 1 hr to reach the hospital

Project A

They travel from places which takes 2-6 hrs of time to reach the hospital.

Project A

Takes 5hrs to reach the hospital

Project A

They arrive via two wheeler vehicles.

Project A

They are carrying helmets inside bcz NO PLACE to keep them safely

Project A

Each person carries it bcz of traffic rules

Project A

Special Case

Project A

In family, 80 year old father having health issues

Project A

In this case, the superintendent waive off the medical charges .

Project A

WAITING HALL

Project A

came to sit in waiting hall bcz of exhaustion by the hospital procedures

Project A

Attendants stands in the queue.

Project A

Patient sits on the benches in the waiting hall

Project A

people keep entering and exiting via one common gate of the waiting hall.

Project A

After 12pm, the crowd is less

Project A

The queue length is less.

Project A

8AM to 12 PM the waiting halls are packed with people

Project A

In such a crowd, there are chances of injuries here.

Project A

TEST

Project A

yellow receipt - urine test

Project A

Pink receipt - blood test

Project A

white receipt -

Project A

Tests cost 25000rs

Project A

CT scan got done today

Project A

Green receipt-

Project A

Reports

Project A

The reports takes time to get to the patients/ attendants.

Project A

The reports of CT scan, MRI, takes time to get to the patient.

Project A

Each and every detail is written in the reports itself

Project A

Meals

Project A

Meals from the truck outside the hospital

Project A

Here 2-3 hrs are wasted

Project A

For water also there is a queue

Project A

Lunch Food= Outside Truck

Project A

Medicine And Pharmacy

Project A

Medicine from central pharmacy located in another building

Project A

Amrit

Project A

Digitalization of govt and privt hospital

Project A

When do they digitize them?

Project A

There are 360 degrees cameras

Project A

There is no instant digitalization system of the patients.

Project A

Billing Counter

Project A

Nephew
young boy

Project A

Lack of clarity on
which counter is
for what purpose

Project A

Attendant (nephew) was standing
in bill counter queue.

Project A

Billing counter analyses the
receipt and keeps it in the records

Project A

There are two
copies of receipt

Project A

Once the nephew was standing
at the bill counter gets to know
they are standing in wrong queue

Project A

They rushed back to stand in the
correct queue which again will
take their time.

Project A

The counter of the queue is not
very clear

Project A

This is done with
sign and time.

Project A

People get
confused

Project A

WHAT PEOPLE CARRY

Project A

Luggage

Project A

Folder of medical
files

Project A

Blood samples

Project A

people coming to the hospital
carry their helmets inside the
hospital

Project A

Attendants carry
bag of clothes

Project A

They are carrying
a discharge slip

Project A

with disease
code ID and other
details

Project A

What is everyone
carrying

Project A

Ward boy Carrying a tray of blood
samples with receipts in another
hand

Project A

They carry clothes for changing
within the hospital

Project A

women carry
clothes for their
children as well.

Project A

Carrying a bag of
clothes

Project A

Holding blood
sample receipt
form

Project A

Receipts
medical papers

Project A

A man carrying a can of water
,maybe 1 litre along with luggage

Project A

Attendants carry the patient's File
filled with reports and paperwork

Project A

was holding and
carrying blood
holder tube and
papers

Project A

They carry extra
clothes and
bedsheets with
them

Project A

Days Spent for Hospital Procedures

Projekta A

Came to the hospital on 15th nov

Projekta A

4 days daily visit for hospital procedures

Projekta A

Patient is admitted now for 15-20 days

Projekta A

on 18 th nov = discharging the patient probably total 5 days

Projekta A

came here on 19th nov total days spent - 3 days in hospital

Projekta A

Total Attendants

Projekta A

4-5 attendants are necessary to bring to the hospital.

Projekta A

5 Attendants came for one patient

Projekta A

Total attendants=5 Entire family

Projekta A

Female 40 yrs old

Projekta A

Male 50 yrs old

Projekta A

Mostly the attendants are the family members

Projekta A

Accompanied by in-laws and Aunt

Projekta A

Total attendants ~4

Projekta A

Doctor's Interview

Projekta A

User Interview -8
Doctor at IIT hospital

Projekta A

MBBS Grad

Projekta A

OPD of medicines

Projekta A

in practice after mbbs degree, they practice everything

Projekta A

In PG there is a specialization

Projekta A

Queue times depends upon the demand of dr.

Projekta A

If demand is high queue length is long

Projekta A

Diabetic patient will have a longer queues

Projekta A

if Dr's timing is 9 to 10 AM patient comes early in the morning

Projekta A

In OPD surgical if a patient needs immediate treatment then the Dr leaves the OPD

Projekta A

After 2 hrs of surgery he will again come back to OPD

Projekta A

Something Scannable

Projekta A

Dr's Handwriting is an issue

Projekta A

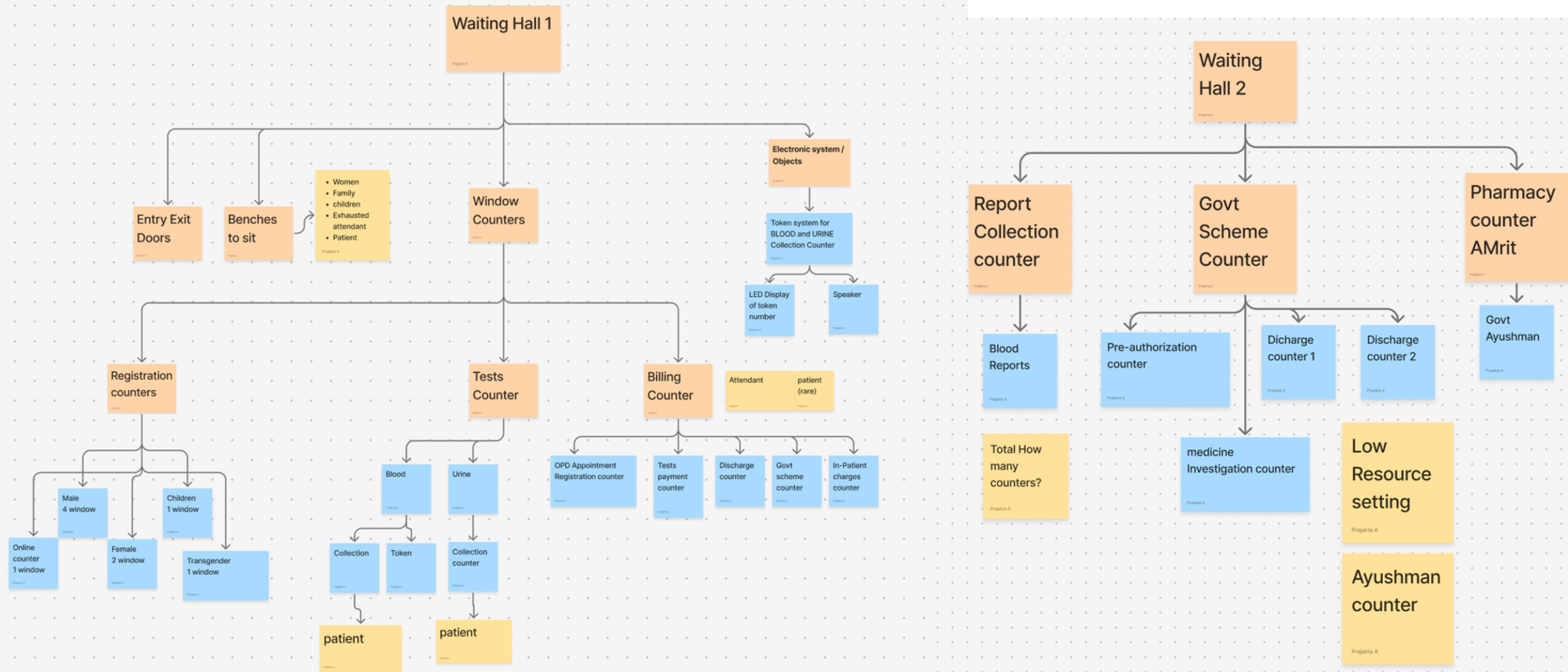
Today one Dr 500 patients

Projekta A

Suggestion : a pen like product to write and tell the follow up procedure

Projekta A

Patient Service Flow and Infrastructure





Ramesh das

Patient

About



58



Rural village
near Barpeta,
Assam



Farmer



Primary school
(Class 5)



Married

Income : Low

Hopeful

Resilient

Trusting

Cooperative

Motivations

Ramesh seeks quality and affordable healthcare for his chronic health issues. He wants to recover quickly so he can return to work and support his family. He values clear communication from doctors and hopes for a respectful experience at the hospital.

Goals

- Get proper diagnosis and affordable treatment
- Navigate the hospital with minimal confusion
- Minimize time and costs spent on travel, stay, and waiting

Pain Points

- Long waiting times and unclear queue process
- Difficulty in understanding signs and instructions (language + literacy barrier)
- Frustration when files get misplaced or he's asked to visit multiple counters
- Physically exhausted from standing in long lines
- Anxiety and confusion when navigating large, crowded hospital spaces

Behaviours

- Relies on verbal instructions from hospital staff or other patients
- Often comes early morning to "get in line"
- Frequently skips meals to wait in queues
- Feels hesitant to ask too many questions

Technical Skills

Basic Mobile Phone



Social Networks



Online Shopping





Sita Das

Patient's Attendant

About



38



Rural village near
Barpeta, Assam



Homemaker



No formal Schooling



Married

Income : Low

Supportive

Caring

Responsible

Resilient

Motivations

Sita is motivated by her desire to ensure her husband receives the best care possible. She wants to understand the treatment process and help him recover. She values kind staff and clear, simple communication.

Goals

- Support her husband emotionally and physically during hospital visits.
- Navigate hospital processes efficiently (registration, tests, pharmacy).
- Ensure she understands discharge instructions and follow-up care.
- Find affordable accommodation and food nearby if needed.

Pain Points

- Overwhelming and confusing hospital procedures.
- Difficulty communicating with staff due to language barriers.
- Lack of comfortable waiting areas.
- Managing her own stress and fatigue while caring for her husband.

Behaviours

- Holds on to all prescription slips, test reports, and documents
- Relies on informal help (asking other attendants, guards, or nurses)
- Carries food and water from home to reduce costs
- Becomes impatient if queue doesn't move or when not informed clearly

Technical Skills

Basic Mobile Phone



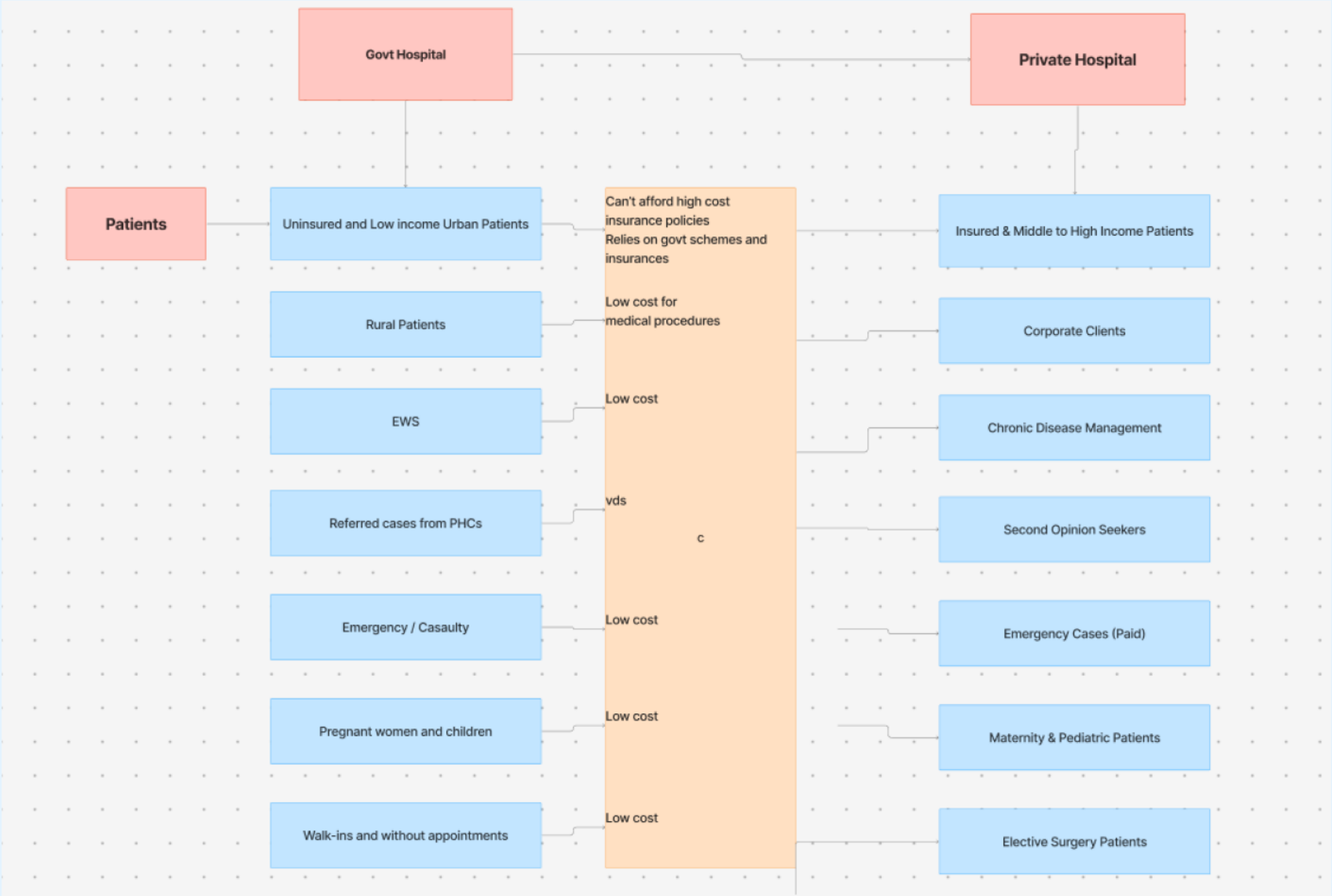
Social Networks



Online Shopping



Aligning Healthcare Standards Across Hospitals



Attendants

Family Members

Family Members

Few will be literate about the procedures

Most of them will be literate about the hospital procedures

Digital/Tech Literacy here is LOW

Digital/Tech Literacy is HIGH

Stress Level if HIGH

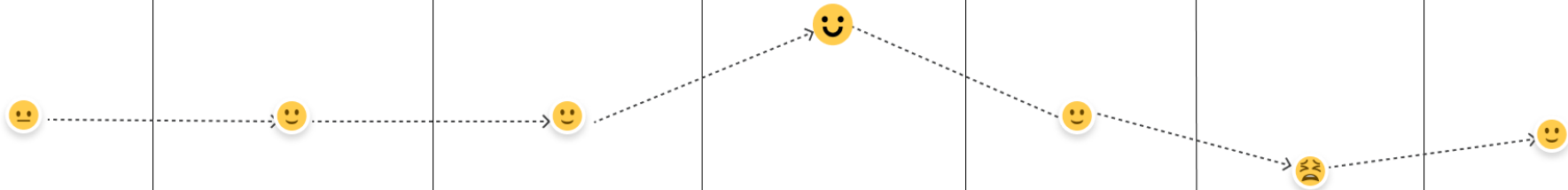
Stress Level is LOW

Low

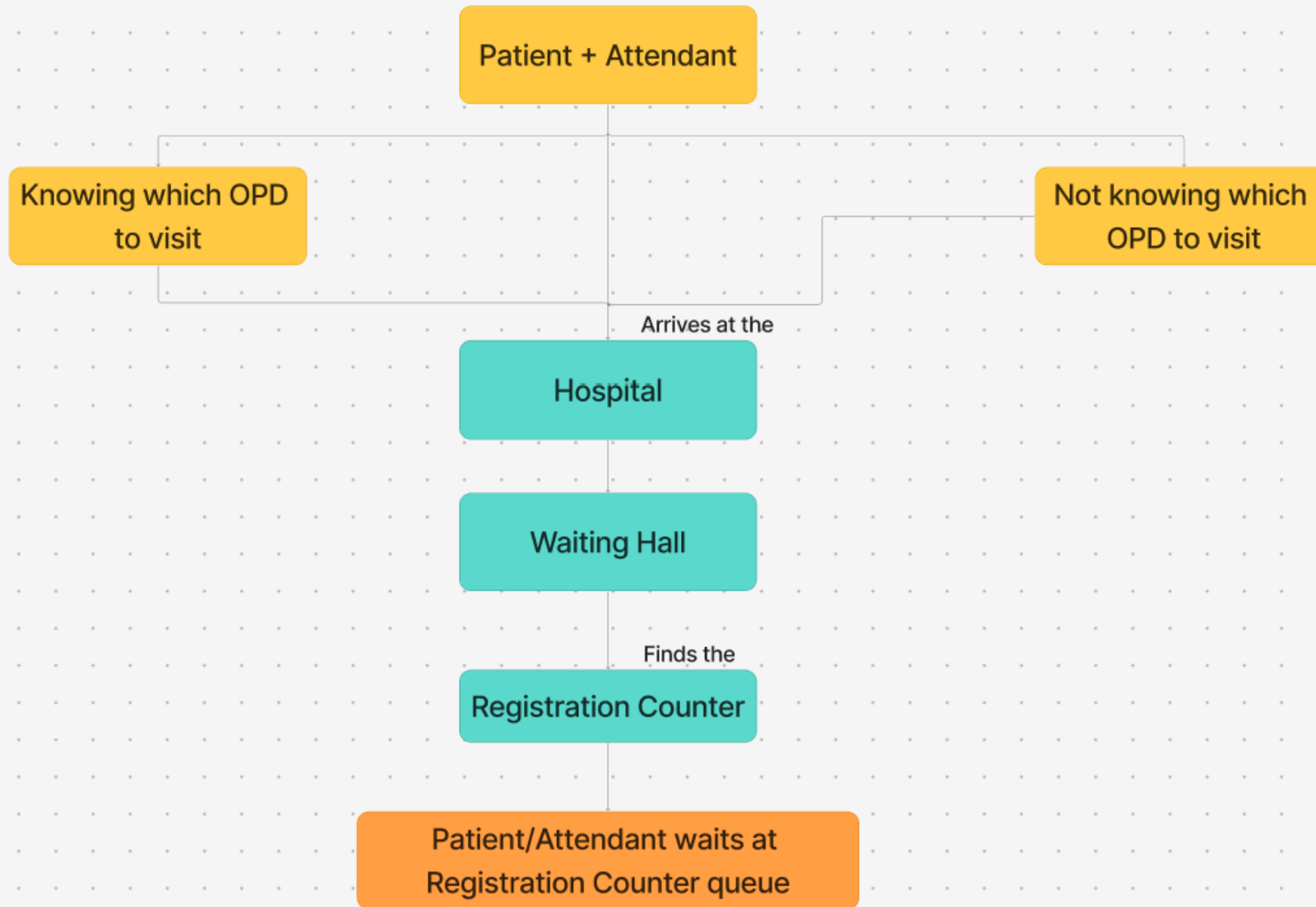
Patients



Goal: Patient and Attendant's Journey

PHASE	Waiting Hall	Registration Counter	OPD		Doctor Consultation	Billing Counter	Test Counter
User Action	Entering the Patient and attendant and reaching the counters	Standing in the Queue	Waits for OPD Screening for the allotted OPD	Enter the Doctor's Cabin for Consultation	User gets to discuss their problems and receives proper medication	Again has to stand in the Queue of Billing Counter	After Payment of Test, wait in the queue of
User Emotion	 A line graph showing the user's emotional journey across the phases. The x-axis represents the phases: Waiting Hall, Registration Counter, OPD (split into two columns), Doctor Consultation, Billing Counter, and Test Counter. The y-axis represents the emotion level, indicated by smiley face icons. The journey starts with a neutral smiley face in the Waiting Hall, remains neutral through the Registration Counter, and then rises to a happy smiley face at the start of the OPD section. It peaks at a very happy smiley face during the Doctor Consultation, then drops to a sad smiley face at the Billing Counter, and finally rises back to a neutral smiley face at the Test Counter. <pre>graph LR; A[Waiting Hall] --> B[Registration Counter]; B --> C[OPD]; C --> D[Doctor Consultation]; D --> E[Billing Counter]; E --> F[Test Counter];</pre>						
User Expectation	Getting the to the Hospital ASAP.	- Getting the case paper and rushing towards OPD - The man behind the counter should help me find the right OPD	Waits for the OPD screening	- Finally getting to discuss their problems - Trustworthy consultations	Good care provided by the doctor after consultation	For tests again has to stand in the queue for the payment of Tests provided after consultation	To get the test done ASAP

Current Journey



Patient/Attendant waits at
Registration Counter queue

Patient/Attendant
receives Two Sheets

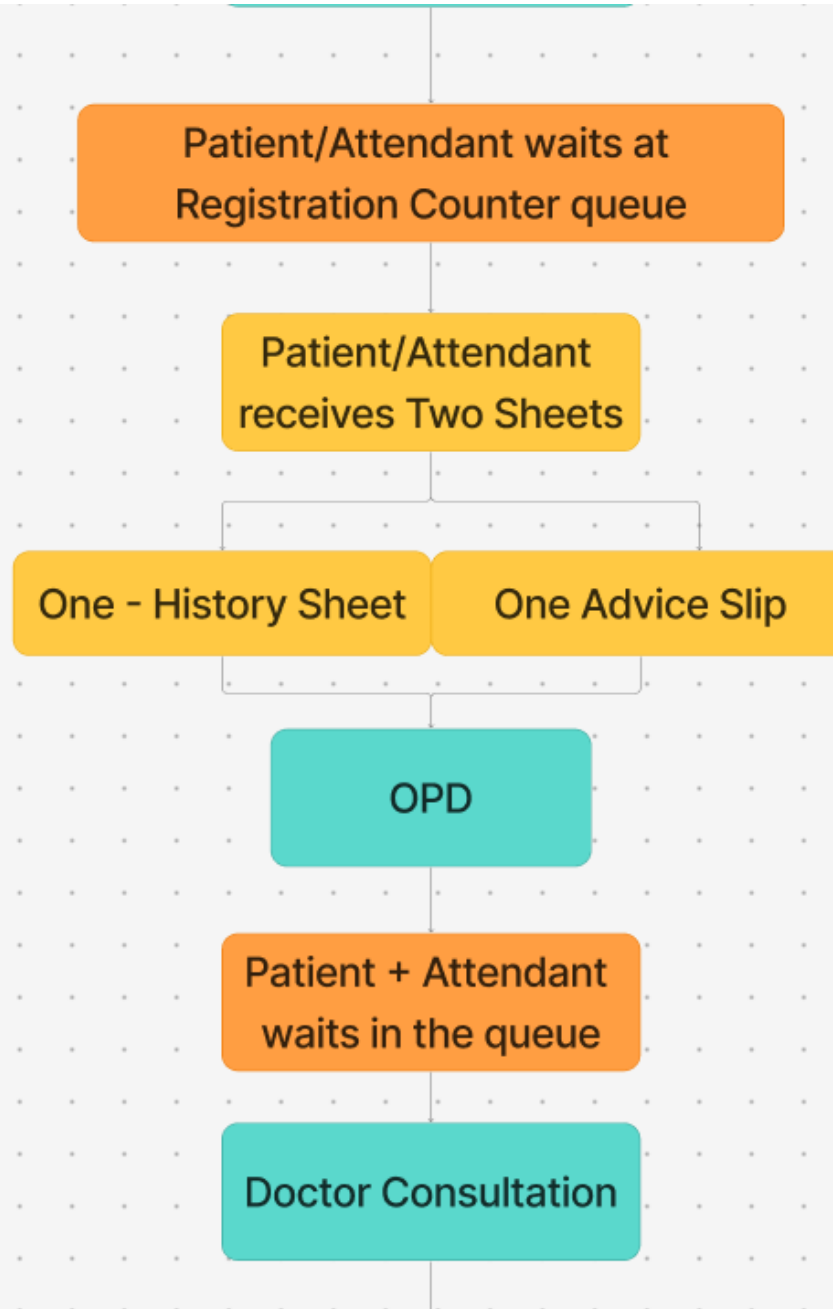
One - History Sheet

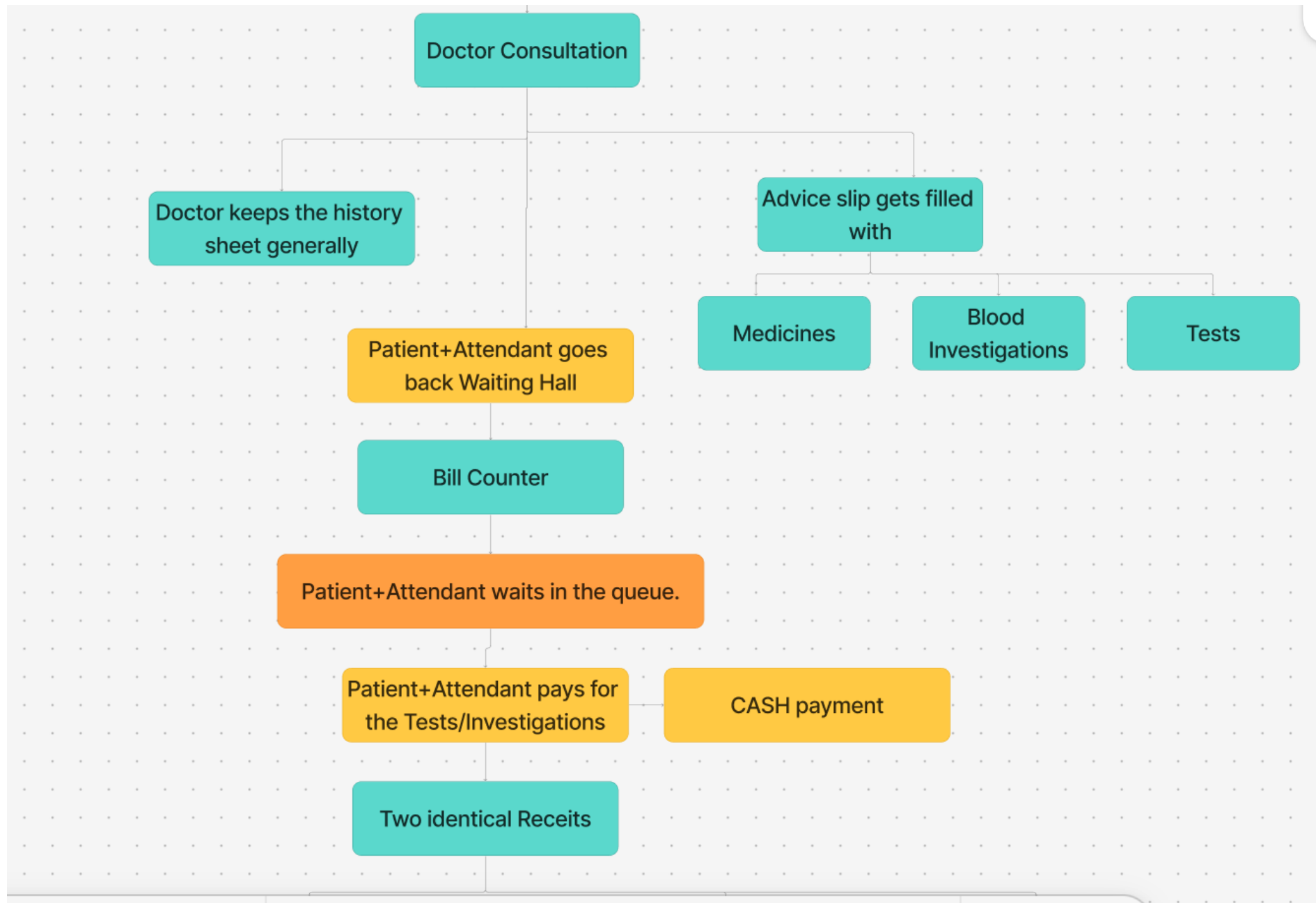
One Advice Slip

OPD

Patient + Attendant
waits in the queue

Doctor Consultation







Two identical Receipts

Blood Test

Patient+Attendant goes
to the Blood Collection...

Patient+Attendant wait in the queue

Here Token system is followed

Patient+Attendant blood wit...

Blood samples are segregated
for respective Labs.

One receipts is taken by the...

One receipts is given to Pati...

with Counter Number, Date,...

Ultrasound

Patient+Attendant goes
to the Ultrasound Coun...

Same Procedure

Xray

Patient+Attendant goes
to the X-Ray Counter/lab

Same Procedure

Pugh Matrix

Criteria	Manual Queue	Token System	Digital Display Queue	AI-Based Dynamic Queue
Waiting Time Reduction	0	+1	+2	+3
Patient Satisfaction	0	+1	+2	+3
Implementation Cost	0	-1	-2	-3
Ease of Use	0	+1	+1	+3
Scalability	0	+1	+2	-2
Staff Training Needed	0	-1	+1	-2
Real-Time Adaptability	0	-1	+2	+3
Data & Analytics	0	-1	+2	+3

MOSCOW

MUST HAVE

- **RFID Card Scanning:** Identifies patients quickly and securely.
- **16x2 I2C LCD Display:** Displays patient details and system messages.
- **Coin Vibration Motors:** Gives tactile feedback to confirm actions.
- **Servo Motor/Token Dispenser:** Automates token dispensing or patient flow.
- **Pay, Print, Cancel Buttons:** Simple, accessible controls for all users.
- **Token Number Generation:** Ensures orderly and fair queuing.
- **Power Backup Support:** Keeps the system functional during power cuts.

SHOULD HAVE

- **Mini Thermal Printer:** Prints physical tokens with patient info.
- **Multi-Language Display:** Supports Assamese, Hindi, and English.
- **Basic Analytics Logging:** Tracks patient traffic for hospital insights.
- **Sound/Buzzer Output:** Adds auditory cues for visually impaired.
- **Digital Clock Integration:** Shows real-time timestamps on tokens..

COULD HAVE

- **Mobile App Notifications:** Sends tokens to patients' phones.
- **Touchscreen Interface:** Replaces buttons with visual icons.
- **Queue Display Screen:** Shows current and upcoming token numbers.
- **Database Sync:** Connects patient data with hospital records.
- **NFC Smartcards:** Speeds up check-in with secure contactless tech.
- **ABHA/Aarogya Setu Integration:** Aligns with India's digital health mission.

WON'T HAVE

- **AI Smart Scheduling:** Too complex for basic queuing.
- **Full EMR Integration:** Depends on complete hospital digitization.
- **Facial Recognition Entry:** High cost and privacy concerns.
- **Cloud-Based Management:** Needs stable internet, often lacking in public hospitals.

Design Brief

Objective:

To design an interactive, user-friendly queue management system that enhances the patient and attendant experience in both government and private hospital settings by reducing confusion, wait times, and inefficiencies.

Target Users:

- Patients visiting OPDs (especially elderly, first-time, and uneducated patients)
- Attendants accompanying patients

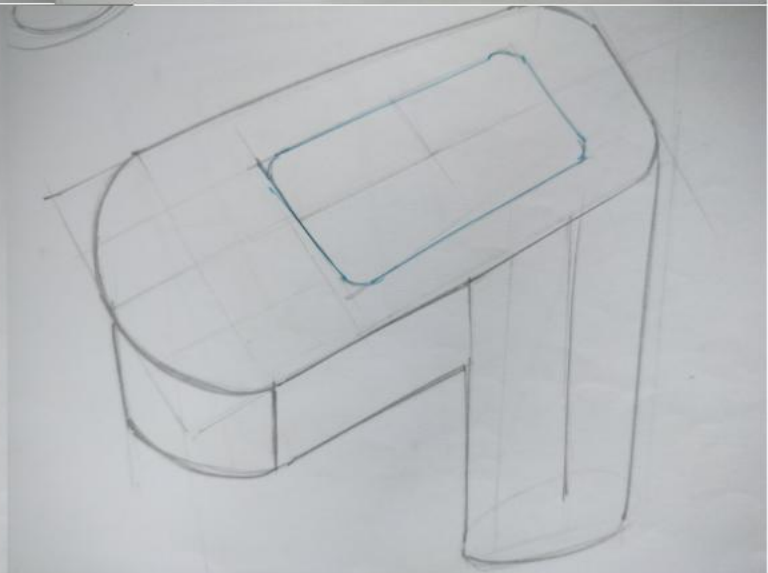
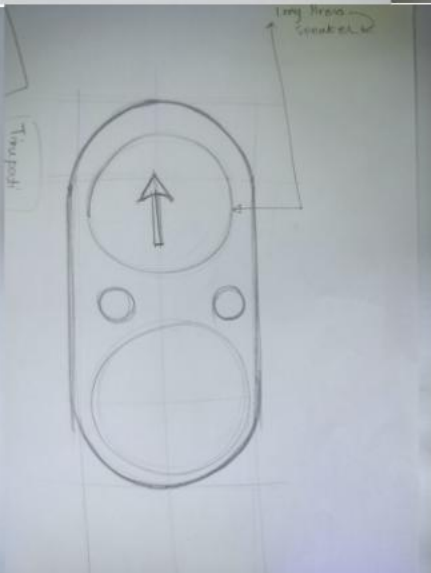
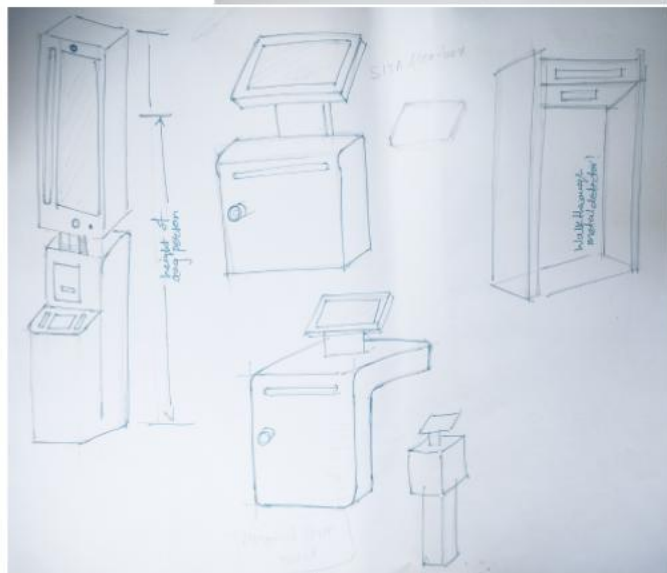
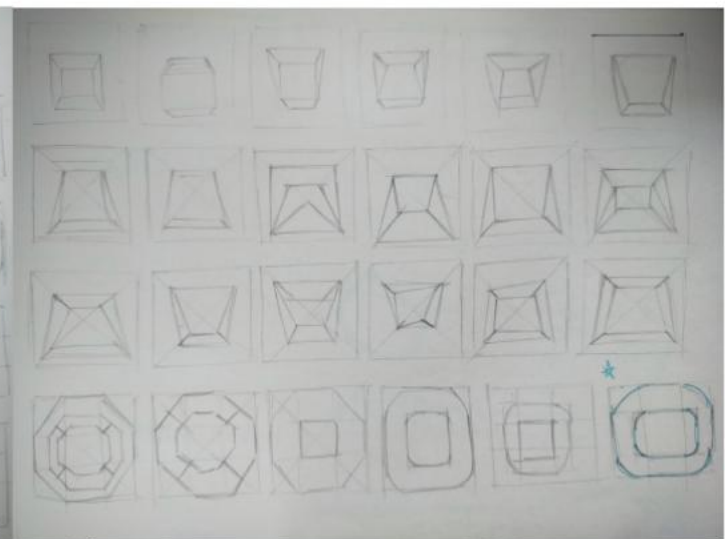
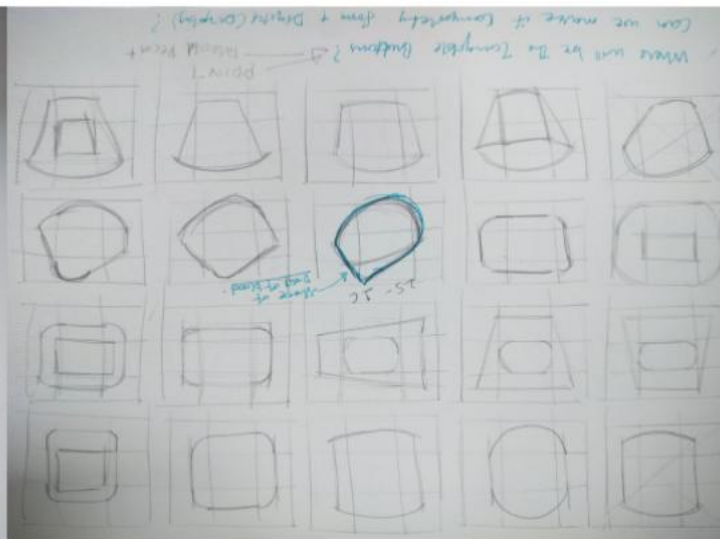
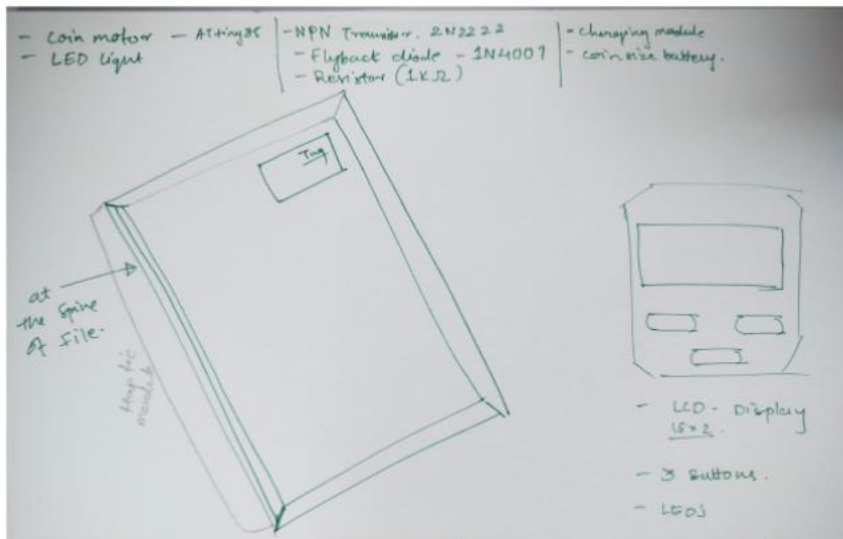
Design Goals:

- Simplify the queue experience for patients and attendants
- Improve communication and feedback during the waiting process
- Make the system accessible, inclusive, and adaptable to low-resource environments
- Propose scalable technology that can evolve with hospital infrastructure

Problem Statement

In many hospitals and service institutions, bill payment counters are congested, slow, and manually managed, leading to long waiting times, miscommunication, and frustration for patients and their attendants. The lack of an organized and automated payment process often results in:

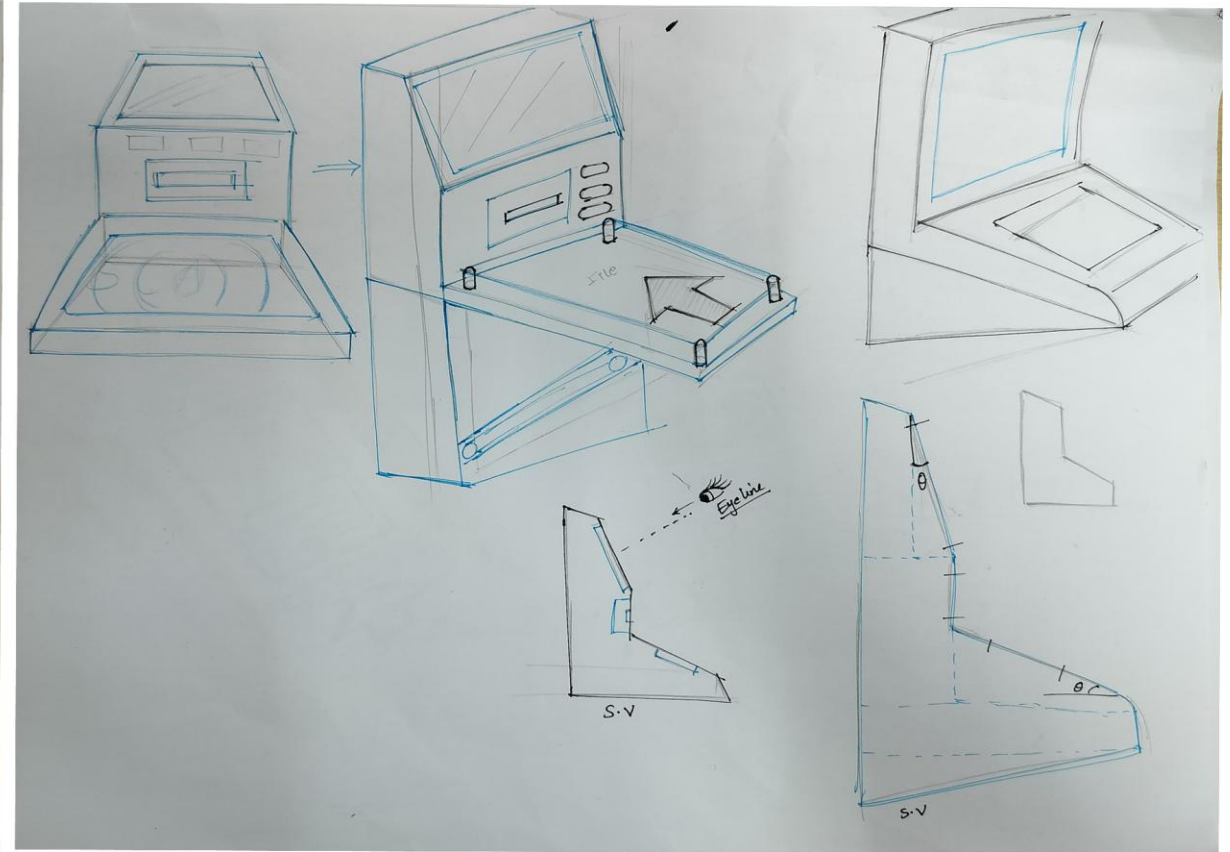
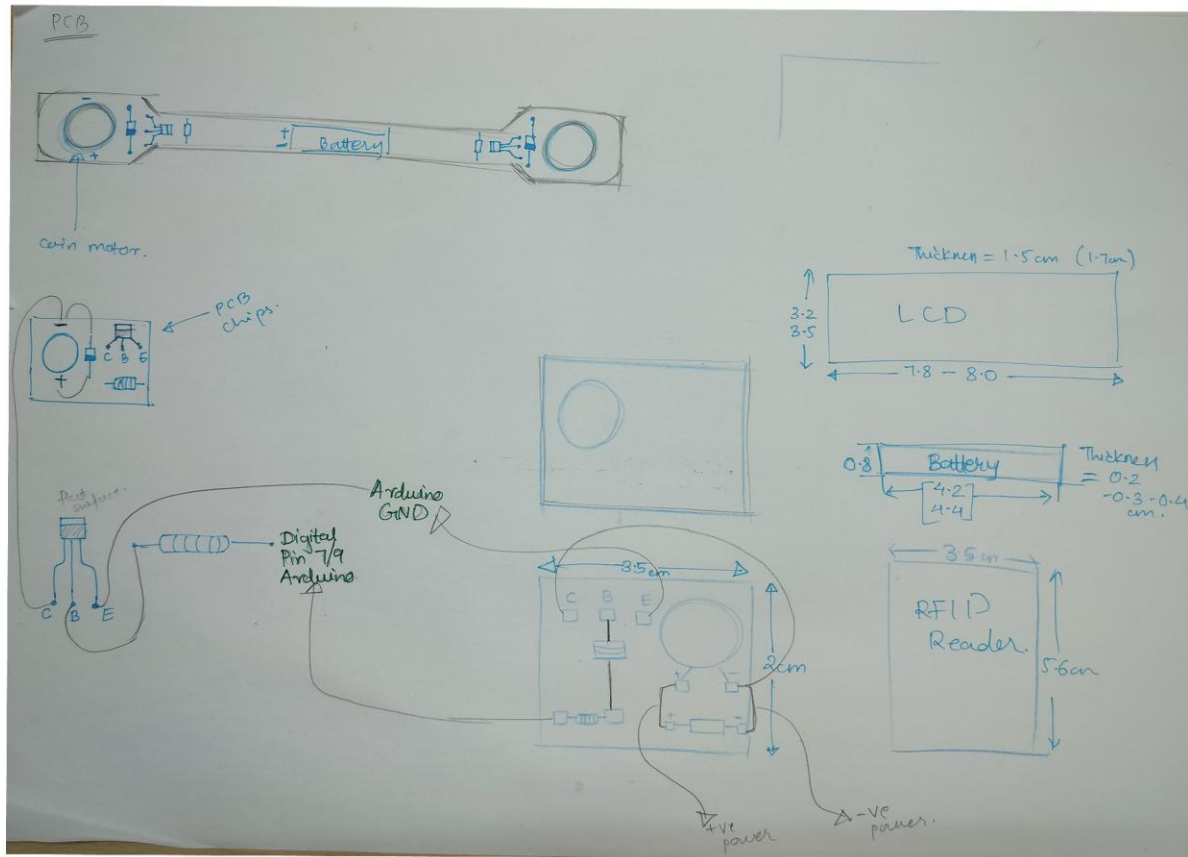
- **Queues forming without a proper token or turn-based system**
- **Delays in treatment or doctor consultations due to billing bottlenecks**
- **Difficulty for elderly or less tech-savvy users to navigate the process**
- **Overburdened staff handling both billing and crowd control simultaneously**



Initial Form Exploration - Ideation Sketches

Concept 1

A small kiosk with a RFID Scanner which scan the tag on the patient's file, LCD Display, Coin motors for haptic feedback and a mini Thermal printer for obtaining patient's token number.



Electronics Components



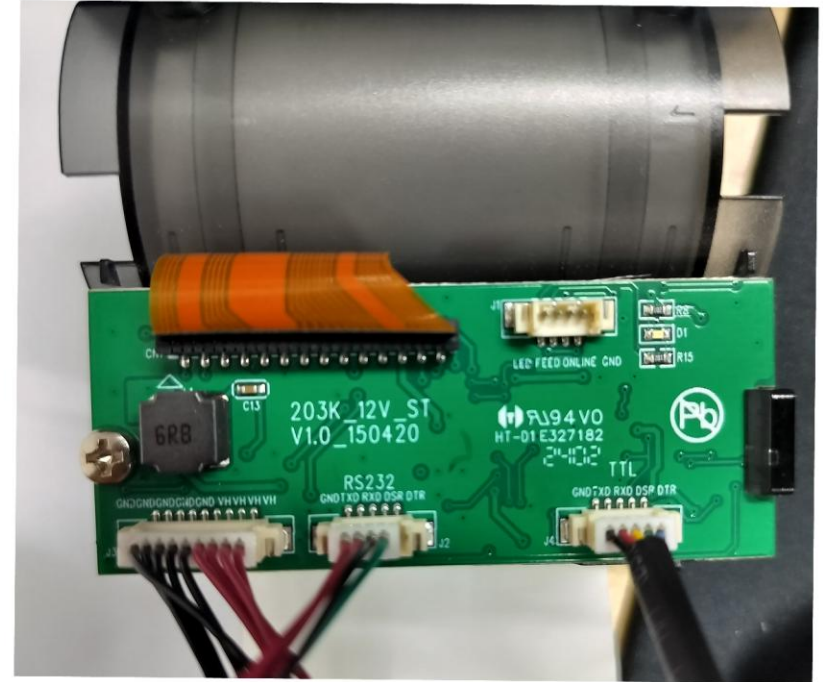
RFID RC522

Add a quick description of each thing, with enough context to understand what's up.



Coin Motor

Keep 'em short and sweet, so they're easy to scan and remember.

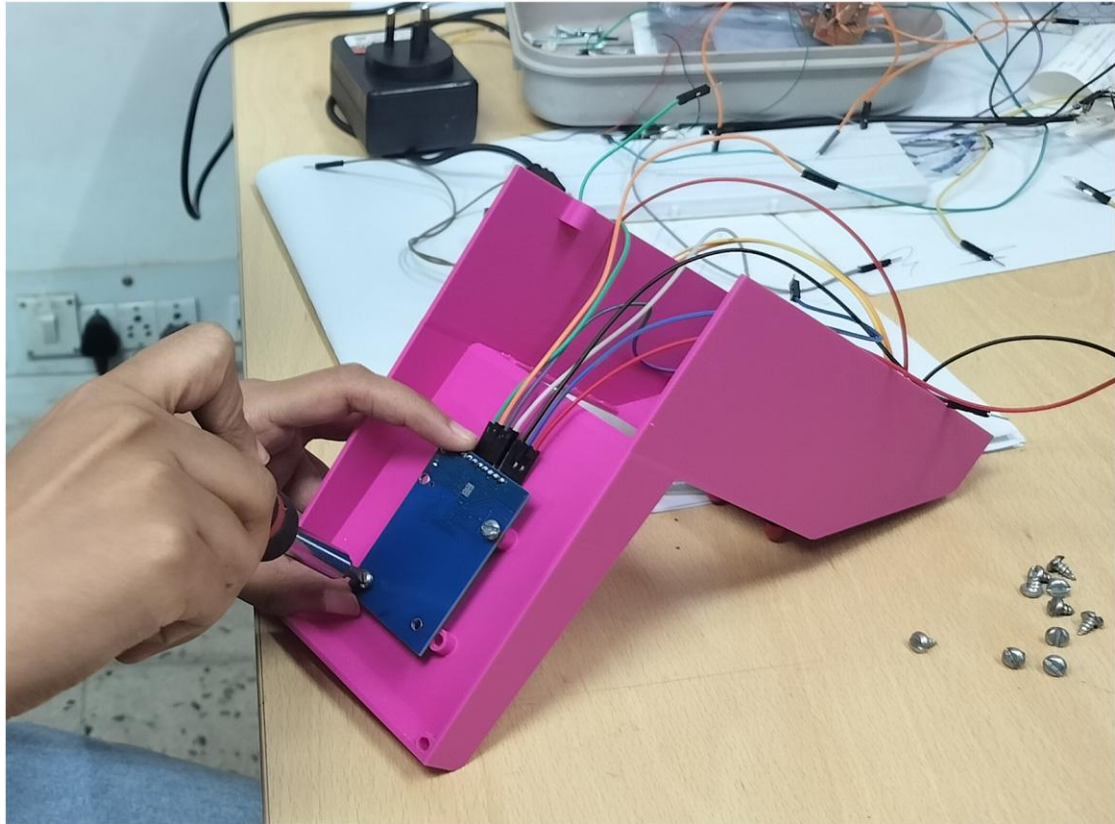


RP203 Mini Thermal Printer

If you've got a bunch, add another row, or use multiple copies of this slide.

Prototype

Phase - 1



RFID Testing

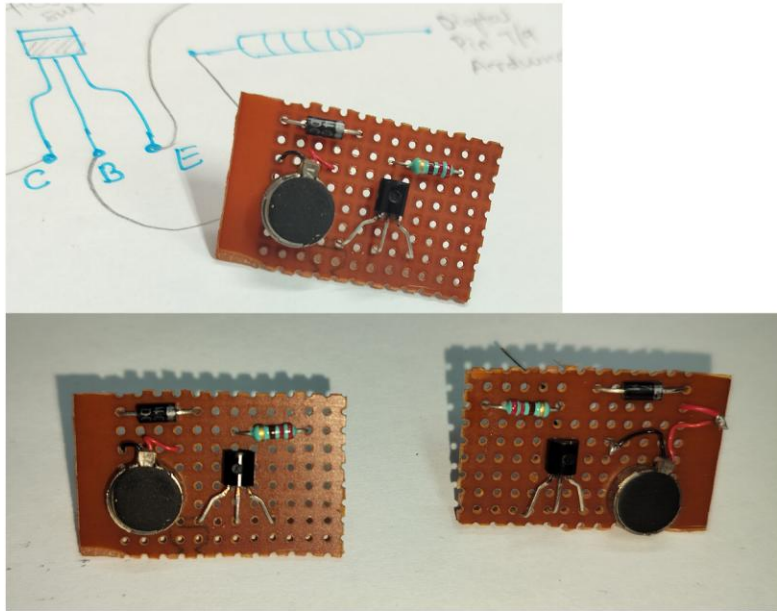
RFID RC522 Testing with the prototype



First Iteration

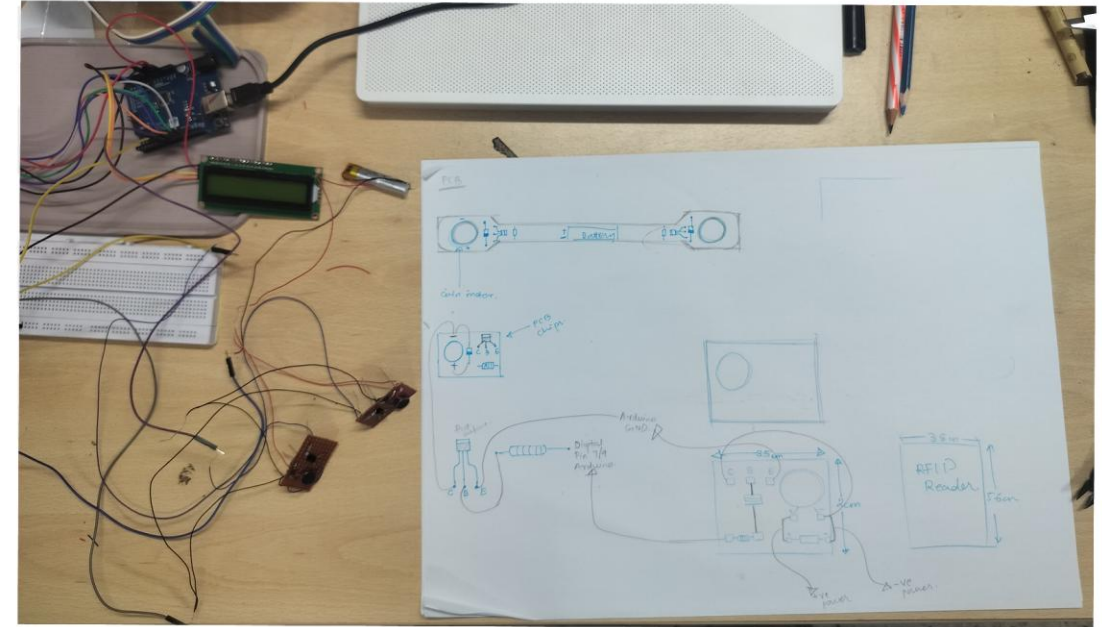
Testing the RFID with the #D printed model

Testing Coin motors



Coin Motor PCB Mounting

A coin motor Vibration module for Haptic Feedback. It is connected with a transistor, Resistor and diode.

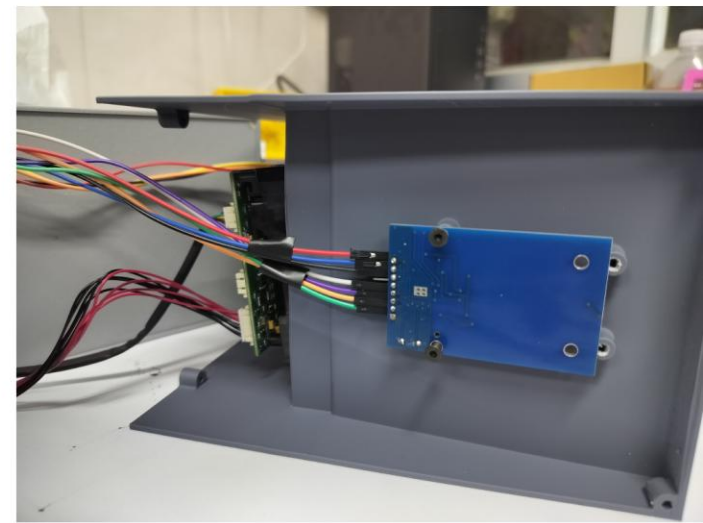
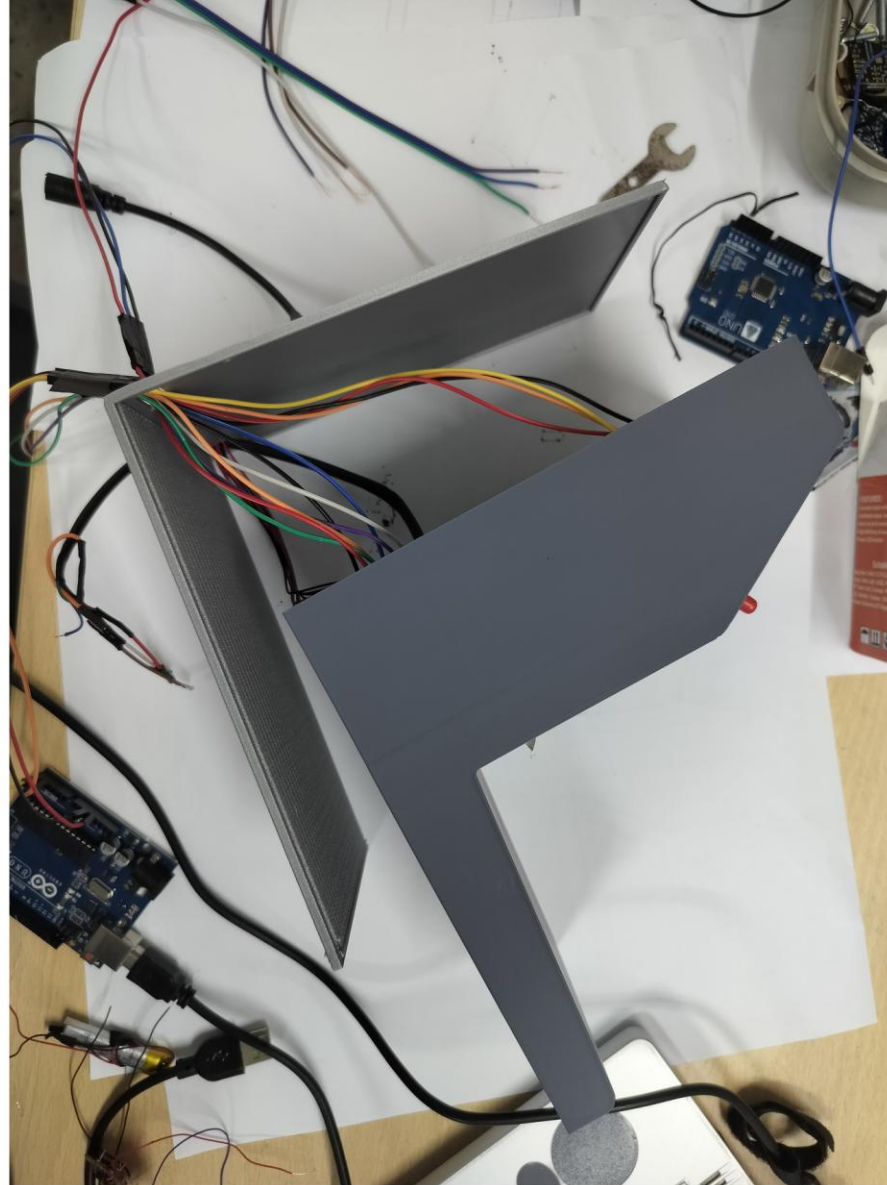


Testing the Coin Motor Module

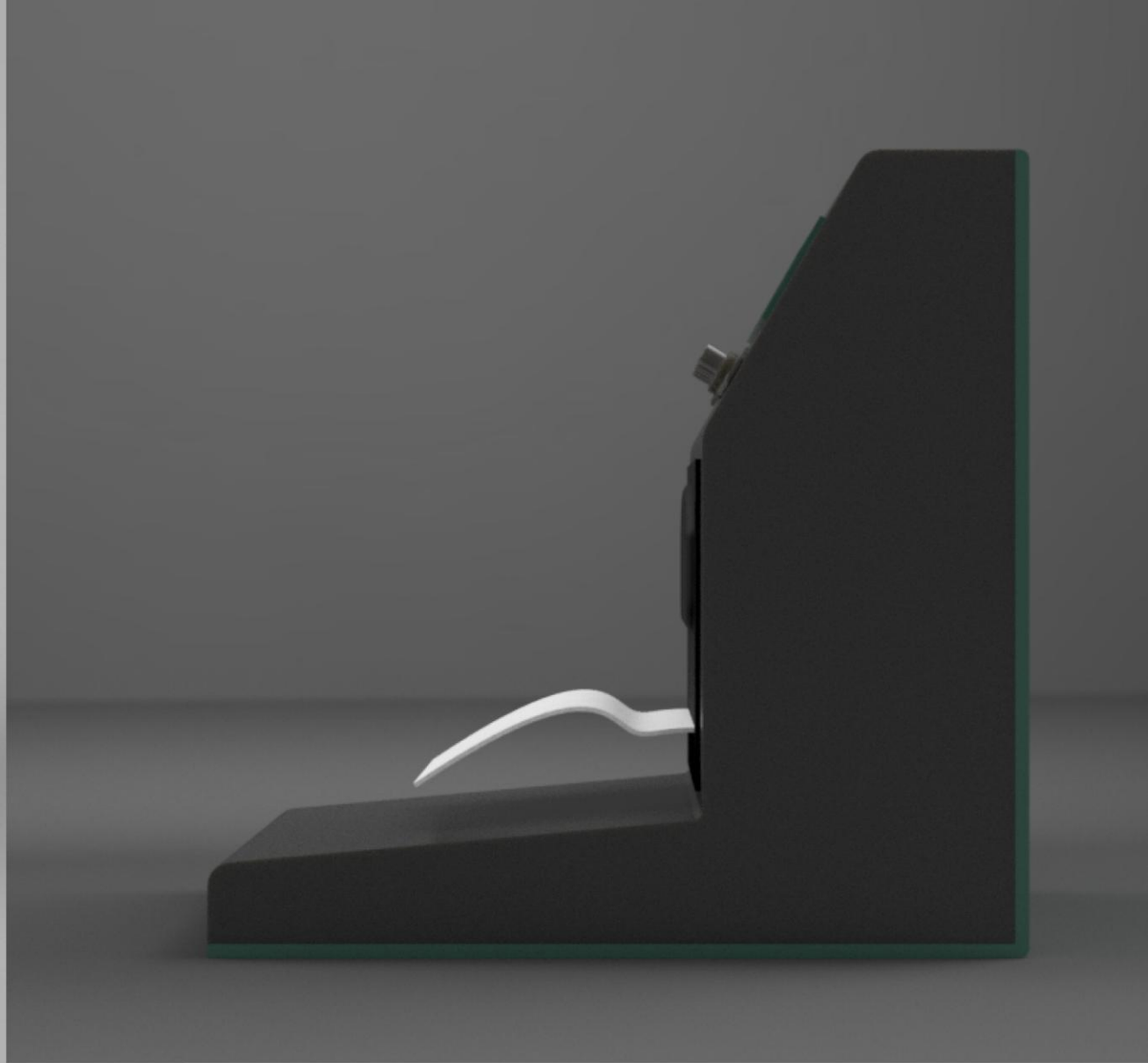
Two coin motors connected in parallel was tested. When the RFID reader scans the tag ,the module vibrates for 3-4 sec.



Iterations of Prototypes



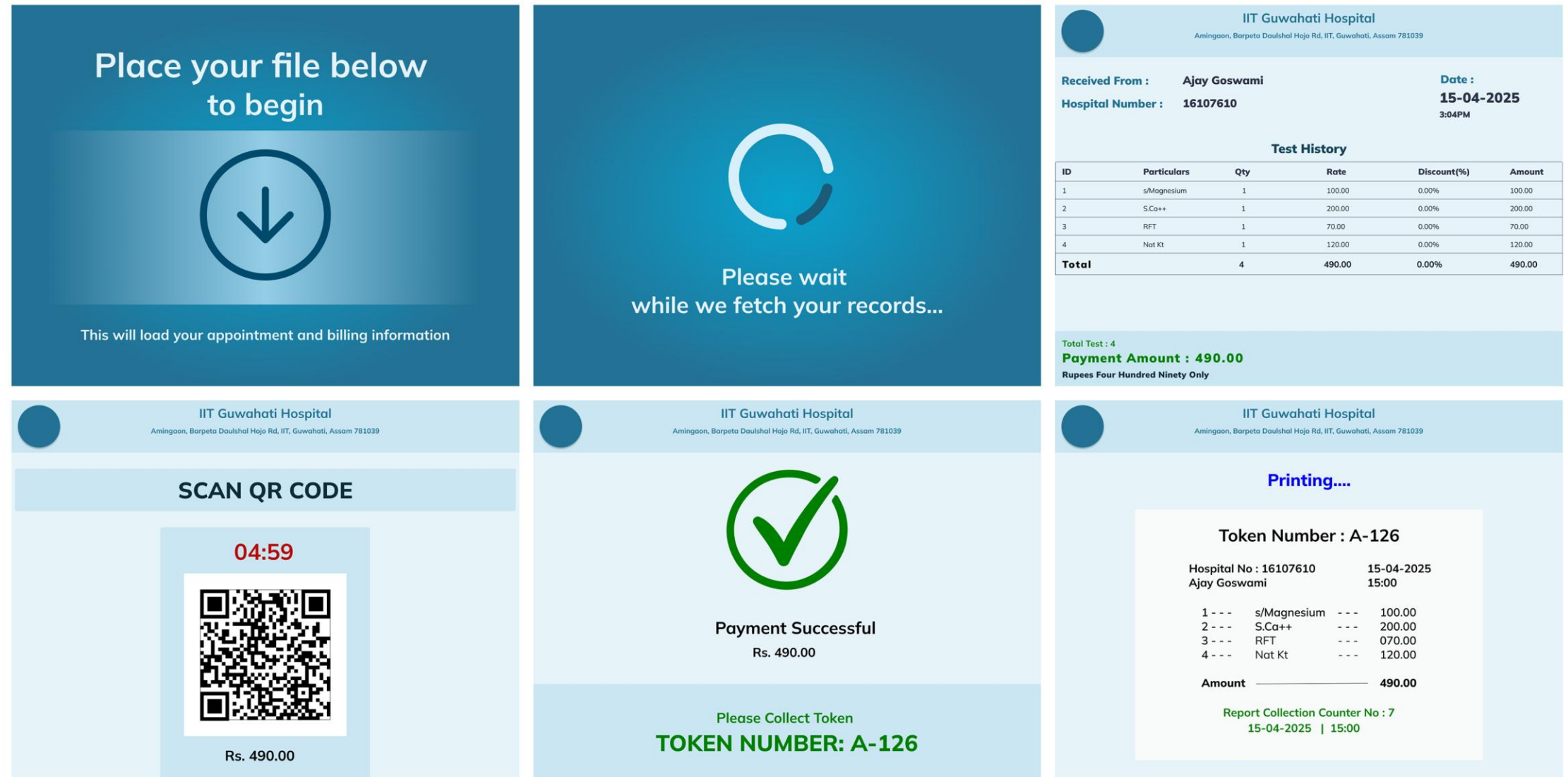
Final Prototype Assembly Pictures



Final Product Renders

Concept 2 : UI Design for kiosk

Adding a Display to this system, for testing 16X2 LCD is used, it can also have a display of diagonal length 5". For this, the prototype can be taken further with a bigger screen having a interface to interact with the user.



Future Aspects

Electronics Upgrade

- Introduce **NFC-enabled smartphone check-in** to remove dependency on physical cards
- Replace RFID cards with **biometric authentication**.
- Connect to cloud dashboards for hospital admins to monitor and manage queues remotely

System Integration

- Integrate with hospital HIS (Health Information System) for seamless patient data access.
- Add patient triage system to prioritize critical cases automatically

Other Improvements

- Multilingual interface + voice instructions for inclusivity
- Smart seating allocation based on queue and needs